

Chapter 2 Study Sheet

Chapter 2 test · Population and Health · the whole chapter, pp. 1–56

Formulas

Natural increase rate

$(\text{CBR} - \text{CDR}) \div 10 = \% \text{ per year}$

CBR 35, CDR 10 → 25 per 1,000 → 2.5%

Total population change

natural increase + immigration – emigration

natural increase alone leaves out migration

Doubling time

$70 \div \text{growth \%}$

the math behind the book's \$100 at 10% → ~7 years. 2.5% → about **28 years**; 2% → 35; 1.17% (India) → about 60

Old-age dependency ratio

people 65+ ÷ people 15–64 × 100

12 million ÷ 40 million → **30**

Total dependency ratio

(ages 0–14 + ages 65+) ÷ ages 15–64 × 100

Replacement level

TFR of **2.1** keeps a population stable without immigration

Physiologic vs. arithmetic

the bigger the gap, the less of the land is farmable

Density

- **Population density:** people vs. land, as if spread evenly.
- **Arithmetic** = people ÷ *all* land. U.S.: 326 million, **86 per sq mi** (33 per sq km).
- Bangladesh **2,962** · Netherlands **1,068** · Japan **869** per sq mi.
- Misleads: hides empty Alaska and the West, and Egypt's desert.
- **Physiologic** = people ÷ *arable* land.
- **Egypt:** 97.6 million; 252 → **6,995** per sq mi; **98%** of people on **3%** of the land (Nile valley and delta).
- **Switzerland:** physiologic = 10 × arithmetic (1 acre in 10 arable).
- **Ukraine:** 44.8 million; 192 → 281 (× 1.46: 1 of every 1.46 acres arable).
- **Agricultural (AP)** = farmers ÷ farmland: rural Bangladesh very high, Iowa very low.
- Middle America and the Caribbean high; South America moderate; India lowest in South Asia and has more farmland per person than China.

Population data

- U.S. census every **10 years**: money and seats in Congress depend on it.
- Undercounted: migrants, racial minorities, doubled-up or homeless families, low-income families.
- Debates: **sampling** vs. counting everyone; the **citizenship question**.
- UN, World Bank, Population Reference Bureau: numbers don't always match; often **estimates**.

Distribution and clusters

- **Distribution:** the pattern people make across space. **Clustered** = close together; **dispersed** = few, spread out.
 - People lived near food; farm technology and shipping loosened that link.
 - **Dot map** (Fig. 2.4: 1 dot = 100,000 people). **Dasymetric map** (Fig. 2.5) uses land cover and terrain.
 - China + India = $\frac{1}{2}$ of humanity (empty Himalayas, China's desert).
1. **East Asia:** almost $\frac{1}{4}$ of humanity, 1.4 billion in China; Huang He and Yangtze ribbons; wheat and rice.
 2. **South Asia:** 1.5 billion; India, Pakistan, Bangladesh, Sri Lanka; Ganges, Indus, Brahmaputra; hemmed in by Himalayas, Afghan/Pakistani mountains, Indian Ocean. India 1.17 vs. China 0.56; India most populous by 2030 (1 in 6).
 3. **Europe:** 725+ million; dense even in mountains; Industrial Revolution first; urban: Germany 77%, UK 83%, France 80%.
 4. **North America:** megalopolis Boston–Washington, D.C. and Quebec City–Windsor; 70+ million; half of Canada; 20% of Americans; $\frac{1}{4}$ the size of Europe's cluster.
- Big three ≈ **4 billion** of 7.7 billion. South America + Africa barely beat India.
 - Dhaka **122,700** · Mumbai **68,400** · NYC **28,717** per sq mi. Rural Bangladesh 3,000–5,000 vs. rural Iowa **55**.

Opening and big ideas

- **Shanghai:** 128-story tower (2,000+ ft); ~22 million; China 1.4 billion, 100+ cities over 1 million.
- Coal → **PM-2.5** (reaches lungs and blood). North China lives **5.5 years** less. Fixes: limit coal, ban imported plastics for burning, limit cars.
- **Demography** = study of population trends.
- **Scale:** downtown → suburbs vanishes on a world map; national averages hide states.
- **Yanゴン:** density + poverty + weak infrastructure = vulnerability; 2008 cyclone, Irrawaddy delta, ~100,000 dead.
- **Luxor:** environment shapes where people live, not just culture.

Malthus and his critics

- **Ehrlich**, *The Population Bomb* (late 1960s): population outpacing food.
- **Malthus**, 1798: food grows **linearly**, population **exponentially**; revised 1803–1826; assumed a country eats what it grows.
- **Wrong because:** countries aren't closed systems (globalization; mercantilism, colonialism, capitalism); crops spread (potato → Ireland, 1700s); Norway 2% arable, imports food; yields rose (seeds, pesticides, fertilizers, irrigation, GMOs).
- **Neo-Malthusians:** focus on population; limits of food, energy and water; farming's environmental costs.

Growth around the world

- **CBR/CDR** = births/deaths per 1,000 per year. NIR leaves out migration. **Negative:** Europe and Russia.
- Soviet growth fell after **1991**. India nearly 3% thirty years before the book; now Africa **2.43%** vs. India **1.17%**; HIV/AIDS.
- Palestinian territories 2.83 · Afghanistan 2.65 · Sudan 2.55 · Yemen 2.52. Iran, Oman, Morocco dropping.
- Limited women's opportunities + men running family life → high growth.
- India below world avg. but above China; Japan shrinking; Thailand negative; Indonesia and Vietnam slowing.
- South America a bit over 1%; Brazil **2.9%** → **0.8%**; Argentina, Chile, Uruguay low.
- Lowest: **global economic core** (U.S., Canada, Europe, Japan, Australia, Uruguay).
- Russia after 1991: health, alcoholism, drugs, male suicide, economy. Ukraine negative.
- More urban → lower natural increase.

World population

2,000 years ago	250 million
1650	500 million
1820	1 billion
1900	1.6 billion
1930	2 billion
1975	4 billion
2000	6.1 billion
2019	7.7 billion

- Doubling time only **39 years** (mid-1980s) → **54** now. \$100 at 10% doubles in ~7 years.
- Life expectancy **30** (1900) → **72** (2016).
- Levels off at **10–11 billion**, 2050–2100. Now ~1.2% a year, 80+ million a year.

Fertility and aging

- **TFR**: average children per woman 15–49. **95+ countries** (41% of people) below 2.1.
- 2016: world **2.4** (the book also says 2.43), Niger **7.2**, South Korea **1.2**. UN: below 2.2 by 2050.
- Drops: Kenya 3.85; China 6.1 → 1.75 in 35 yrs, 1.5 (2010), 1.62 (2016); Iran 6.8 (1980) → 1.7 (2016).
- **Old-age dependency**: Europe 29.9 → 47 by 2050; sub-Saharan Africa 5.7. Child dependency: Africa 74 vs. Europe 23.
- Aging = pensions and care, fewer young taxpayers. Fixes: **raise TFR** or **immigration**.
- **Japan**: 128.06M (2008) → 126.8M (2017) → ~100M (2050, ~20%); discourages immigration; 98%+ Japanese.
- Raising births: Sweden, Russia (leave, daycare); Japan (men's housework). Limited success.
- Still growing because of India, Indonesia, Bangladesh, Pakistan, Nigeria.

India

- 29 states + 7 union territories. North grows fast; south and west slow (female literacy, land ownership, health care, birth control).
- **1952**: first population program in the world.
- **1976**: forced sterilization of men with 3+ children; quotas; Maharashtra **3.7 million**; riots; **6 million** men in all.
- Now: states set policy (Maharashtra pays newlyweds to wait 2 years); birth control for women, injectables, posters, village clinics.
- **Contraceptive prevalence** (women 15–49): India **55.1%**; Andhra Pradesh 69.8, Maharashtra 63.5, Bihar 26.0, Meghalaya 21.1.
- Lesson: a world map hides what's inside a country.

Religion and growth

- Catholic Church and some conservative Islam oppose contraception. **Northern Nigeria** grows faster than the south.
- But the Catholic heartland grows slowly; the **Philippines** grows fast.
- **Bordeaux** (de Blij, from Dakar): "Where are all the children?" Europe is aging.
- Saudi Arabia ~2%; Indonesia's program from **1970** (coercion + inducement) → 1.6% (2000), 1.1% (2016).

Pyramids



	Expansive	Stationary (chimney)	Constrictive
Births	High TFR	Low, ~2.1 or less	Very low
Deaths	High infant deaths, short lives	Long lives	Long lives; deaths rising
Growth	Fast	Slow / stable	Shrinking
Income	Lower	Higher	Wealthy, aging
Burden	Children (Africa 74)	Rising old-age	Few young taxpayers
DTM	2	4	5
e.g.	Niger, Guatemala	France, Italy, Sweden	Japan

- Males left, females right; 5-year groups, youngest at bottom; **check the top bracket**.
- **Lower-income**: ages 0–14 over **40%**, top three groups under **10%**. Pakistan, Yemen, Guatemala, Cameroon, Laos.
- **Composition** = makeup by age, sex, marital status, education.

Demographic transition model

Stage	Births	Deaths	Growth	Examples and causes
1	High	High, swinging	Low	Most of history. Black Death , 1300s: Crimea → Sicily → north on rats' fleas (contagious diffusion); killed ¼–½; Britain ~4M → ~2M. Famine and war.
2	High	Falling fast	Very high	Europe after ~1750: food, sanitation, soap, medicine. Death rate ~35 (Europe, before 1750) → ~16 (Britain, 1850). Today: Afghanistan, Palestine (vaccines).
3	Falling fast	Falling	Low, still growing	Britain ~1870 through the world wars: women's opportunities, later marriage, fewer child deaths. Brazil (contraceptives); Mali (schooled girls have ~half as many children).
4	Low	Low	Stable or slowing	Britain after 1950: contraception, legal abortion. U.S., U.K. today.
5	Very low	Rising	Falling, negative	Japan, Russia . Zero population growth in ~50 years → a stationary population level (SPL).

Based on western Europe after the Industrial Revolution. Late-1980s World Bank SPL predictions (U.S. 276M by 2035, Brazil 353M by 2070, Mexico 254M by 2075, China 1.4B by 2090, India 1.6B by 2150) were unrealistic: China passed 1.2B in 1994, India 1B in 1998. Newer: China **1.45B in 2030**, India **1.7B in 2060**.

Common traps

- Rates are **per 1,000**: divide by 10 for a percent (25 per 1,000 = 2.5%, not 25%).
- **Stage 2 vs. 3**: if births have started falling, it's stage 3.
- **Natural increase leaves out migration**: a negative NIR country can still grow through immigration.
- **Arithmetic vs. physiologic**: pressure on farmland is physiologic.
- **¼ vs. ½**: East Asia is ~¼ of humanity; China + India together are ½.
- **55 vs. 86**: rural Iowa is 55 per sq mi; the U.S. average is 86.
- **3% vs. 2%**: Egypt's people live on 3% of the land; Norway's land is 2% arable.
- **1952 vs. 1976**: India's first program vs. the sterilization drive.
- **2.4 vs. 2.43**: the book gives both for the 2016 world TFR.
- **Over 40%**: the text says ages 0–14 are over 40% in most lower-income countries; use that, not the chart.
- **Religion isn't destiny**: the Catholic heartland grows slowly.

Health and access (2.3)

- **Medical geography:** disease patterns, access to care, and how inequality and local social norms shape spread.
- Health depends on **sanitation** and **access to care**. Access is geographic: **distance** to clinics plus **transportation infrastructure**.
- Social context: HIV/AIDS spread (attitudes to condoms, mobility, income); **obesity** where cheap processed food dominates (heart disease, diabetes).
- Highest health spending per person: **North America, Europe, Japan, Australia and New Zealand** (Fig. 2.19) = lower infant mortality, longer lives. Spending goes to premature babies and the frail elderly (over 85).
- **Mothers' Index** (Save the Children, 13 indicators): income and care matter. **99%** of newborn deaths and **98%** of maternal deaths are in lower-income countries. Conflict drags it down: Iraq, Afghanistan, Syria, Sierra Leone.

Child death rates (per 1,000 live births)

- **Infant mortality rate (IMR):** chance of dying before age 1.
- **Child mortality rate (CMR):** before age 5 (p. 34; p. 38 says ages 1–5).
- **Newborn death rate:** first month. Rich countries: premature and low-birthweight babies. Poor countries: diarrhea and infections cause **half**.
- Causes of high IMR: mother's health, prenatal care, sanitation; mothers with many babies are often poorly fed and schooled; women eat last; **diarrhea and malnutrition** are the top killers.
- ~**30%** lack clean water; **6 in 10** lack hygienic waste disposal; up to **4 billion** lack basic sanitation.
- **Kwashiorkor** = too little protein; **marasmus** = too little protein and calories. In some countries **over 1 in 5** children die between ages 1 and 5.

Infant mortality: numbers and scale

- Lowest: **Japan 1.9** (126M people), Singapore 2.2, Sweden 2.3.
- 2017: five countries at **70+** (CAR, Sierra Leone, Somalia, Chad, DR Congo); **Central African Republic 87.6**, highest ("1 in 8" in the book). 11 countries at 70+ in 2012: falling worldwide.
- Fig. 2.21 caption: high IMR matches **high birth rates** (Fig. 2.7).
- Inside countries: **South Africa 48** on average; whites near Europe's average, black Africans near Africa's.
- **U.S.:** highest in the South, lowest in the Northeast (Fig. 2.22); the whole U.S. is in the world map's lowest class. 2015: African Americans **11.7**, average **6**, non-Hispanic whites **4.8**.
- Prenatal care in the first trimester: **82.3%** of white vs. **66.5%** of African American mothers. Smoking in pregnancy: **10.5%** vs. 6%. SIDS among American Indian and Alaska Native babies: **2.3×** the white rate.

Life expectancy

- Average years a person is expected to live. It's an **average**: low figures mostly reflect **infant deaths**.
- **Japan highest** (may reach **106 by 2300**); Africa lowest. HIV/AIDS pushed some below **40**; lowest now just over **50**. Japan lives ~30 years longer than Afghanistan or Nigeria.
- World (start of the 2000s): women **68**, men **64**. Women outlive men by ~4 years (Europe, East Asia), 3 (sub-Saharan Africa), 6 (North America), 7 (South America), **11 (Russia)**.
- Russian men: **68** → **60** by 2008; **25%** die before 55 (alcoholism, suicide).
- On a pyramid, a wider female top bracket = women live longer.

Kinds of disease

- **Infectious** (~**65%** of diseases): parasites multiply in the body. **Chronic / degenerative:** old age (heart disease, cancer, diabetes). **Genetic:** sickle-cell, hemophilia, cystic fibrosis; cluster in places and groups.
- **Endemic** = small area · **epidemic** = large region · **pandemic** = global.
- **Vector** = carried by a go-between (mosquito, fly). **Nonvector** = direct contact (HIV/AIDS, flu, ebola).
- **Epidemiological transition:** as growth goes from high to stable, deaths shift from **infectious diseases of the young** to **chronic diseases of the old**. AP stage names: 1 pestilence and famine · 2 receding pandemics · 3 degenerative · 4 delayed degenerative · 5 reemergence (measles, HIV, ebola).

Malaria and vectored diseases

- Mosquito bites an infected **host** → parasites multiply in it → next bite infects a new host.
- **>1 million** deaths a year; 300–600 million infected; most victims **age 5 or under**; killed more people than any disease in history. Tropics (moisture), not high latitudes, elevations or dry areas (Fig. 2.24).
- Fighting it: insecticide **nets** (mosquitoes bite at night); **genetically engineered mosquitoes**.
- **Yellow fever:** vaccine can give immunity for 10 years; endemic in tropical Africa and South America. Dengue. **Sleeping sickness:** tsetse fly, from West Africa ~1400.
- **Measles**, 2019: 50+ children in Washington and Oregon; nearly ¼ of Washington kindergartners unvaccinated (place-specific).

What health numbers tell you (link them to 2.1 and 2.2)

- **High IMR** → high birth rate and TFR, stage 2, expansive pyramid, low health spending, women less schooled (Mali, India's north).
- **Falling infant deaths** → families need fewer births → stage 3.
- **Low life expectancy** → mostly babies dying, which also makes a narrow pyramid top.
- **Wider female top bracket** → women outlive men (Russia: by 11 years).
- **Scale:** country averages hide groups (U.S. 11.7 vs. 4.8; South Africa's groups), like Egypt's 98% on 3% and India's north vs. south.
- **Hearths and diffusion:** HIV from Africa, flu from southeastern China, opioids from West Virginia; the Black Death spread by contact (2.2).
- **Russia and Japan** are both stage 5: Japan from long lives and few births; Russia from crisis (men's deaths).
- **The core** (North America, Europe, Japan, Australia) spends most on health and has the lowest growth.

HIV/AIDS

- Nonvectored: sex, then needle sharing.
Hearth: Africa. 1980: ~200,000 infected; 2018: **36.9 million**; **77.3 million** ever. Falling since 2001; surging in eastern Europe and Central Asia.
- 2017, ages 15–49: **Eswatini 27%** (book: "Eswanti", also Swaziland), Lesotho 23.8%, Botswana ~23%, South Africa 18.8%. ~1 million AIDS deaths in 2017. Gould: "continent in catastrophe."
- **Pyramids become chimneys** (South Africa 2035, Fig. 2.25): AIDS kills young adults.
- Elsewhere: China 820,000+ (2018), India >2M, U.S. >1M, Middle and South America ~2M, Southeast Asia up to 5.2M. Undercounted: no early symptoms, stigma.
- **AIDS orphans:** 11.5M → 15M (2001–03); **13.4M** in 2015, 80% in sub-Saharan Africa. Girls leave school to care for sick relatives (Robson, Zimbabwe; Kenya field note).
- Progress: **antiretrovirals** (17M with access, 2016; deaths down 50%+ from 2005). **Uganda:** condoms campaign. Life expectancy: Botswana and Swaziland **34 → 67**; Zimbabwe **36 → 61**. Children with HIV down **47%** since 2010.

Other nonvectored

- **Ebola:** from monkeys and fruit bats; West Africa 2014 killed **11,000+**; DR Congo almost every year.
- **MERS:** less deadly; U.S. by 2014, Korea 2015.
- **Flu:** most common; pigs catch it from birds; source often **southeastern China**. 1918–19: **50–100 million** dead; 2009–10 swine flu ~500,000.

Chronic diseases and addiction

- U.S. 100 years ago: TB, pneumonia, diarrhea, heart disease. TB + pneumonia: **20% → ~2%** of deaths.
- **U.S. 2017** (% of deaths): heart **23.0**, cancer **21.3**, accidents 6.0, chronic lower respiratory 5.7, stroke 5.2, Alzheimer's 4.3, diabetes 3.0, flu and pneumonia 2.0.
- Unknown future risks: new lifestyles, chemicals, preservatives, artificial flavoring, treated water.
- Alcohol: **200M+** addicted; highest **eastern Europe and Russia**; lowest Middle East and sub-Saharan Africa. Cocaine: U.K., Spain, U.S. Amphetamines: Philippines, El Salvador, Australia. Drugs are used far from where they're made.
- **Opioids:** hearth **West Virginia** (Fig. 2.28); **42,000+** deaths in 2016; emergency in 2017; ~**80%** of heroin users started with prescription opioids; highest among non-Hispanic whites.

Population policies (2.4)

- Aim at **growth** or **ethnic makeup**. Three kinds:
- **Expansive** = more births (AP: pronatalist). Used by aging stage 4–5 countries with stationary or constrictive pyramids.
- **Eugenic** = favor one group by discouraging others from having children.
- **Restrictive** = lower natural increase (AP: antinatalist). Used where pyramids are expansive.
- **Word trap:** expansive *pyramid* (wide base, fast growth) is the opposite situation from an expansive *policy*.

Expansive and eugenic

- Past: Soviet Union, Mao's China. Now: tax incentives and family services in Europe and the former Soviet Union.
- **Ulyanovsk, Russia:** Day of Conception, Sept 12 since 2005; **87 births** on June 12, 2008 (~4× a normal day); births **+19.5%**, 2005–2011.
- **Sweden** (1980s, with France): cash, tax breaks, leave, flexibility up to 8 years → mini-boom; economy slowed: class of **130,000** (born 1991) vs. **75,000** (1994); TFR **1.5** by the end of the 1990s, **1.9** now (3rd in Europe). 2015: biggest rise in 70 years, with record **immigration**.
- China now pays highly educated women to have more children.
- **Eugenic:** Nazi Germany; U.S. accused before the 1960s; forced sterilization of LGBT people; Japan's court (Jan 2019) upheld sterilization for gender change.

China's one-child policy

- Restrictive, **1979:** fines, lost school and housing privileges; **minorities exempt**.
- Growth: **3%** (1970s) → **1.2%** (mid-1980s) → **0.5%** today (2.1 says 0.56%).
- Unintended: more abortions, **female infanticide**, abandoned girls (many adopted in the U.S. and Canada); gender and age imbalance.
- 1990s relaxed: rural first child a girl, or both parents only children. **2013:** two children if one parent is an only child. Births didn't rebound; 2050 pyramid is **constrictive** (Fig. 2.30).

One stage, every measure (the whole chapter)

Stage	Pyramid	Health	Main deaths	Burden	Typical policy
1	Expansive	Short lives	Plague, famine	—	—
2	Expansive	High IMR, short lives	Infectious, young	Children (Africa 74 per 100)	Restrictive
3	Expansive, narrowing base	IMR falling	Shifting	Children, shrinking share	Keep births falling
4	Stationary	Low IMR, long lives	Chronic, old	Retirees (Europe 29.9 → 47)	Expansive or immigration
5	Constrictive	Japan IMR 1.9, may reach 106	Chronic, old	Too few young taxpayers	Expansive or immigration

Chapter traps

- **Expansive pyramid ≠ expansive policy:** the policy is used where births are too *low*.
- **Two chimneys:** stationary from low births and long lives; AIDS chimney from young-adult deaths.
- **Life expectancy is an average:** low mostly because babies die, not because adults die young.
- **High CDR ≠ poor health:** an old population has more deaths.
- **Endemic** in the book = small area, not "everywhere."
- **Rate vs. count:** Eswatini's 27% is a share of people; Southeast Asia's 5.2 million is a count. Compare like with like.
- **"Limited success" (2.2) vs. gains (2.4):** Sweden's rebound faded with the economy and still sits under 2.1.
- **Total change** = natural increase + immigration – emigration.